ALA-2

Topic- Keywords and Identifiers

Definition

⟐ Keyword

A keyword is a **reserved word in a programming language** that has a predefined meaning and is used to perform a specific function. Keywords are part of the language syntax and **cannot be used as identifiers** such as variable names or function names.

⟐ Identifier (Definition)

An identifier is a **name given to a variable, function, class, or any other object** in a program. It is used to identify these elements uniquely and is **created by the programmer** following certain rules.

Syntax

Examples of Keyword Syntax:

```
# if keyword
if condition:
    statement

# for keyword
for variable in sequence:
    statement

# def keyword
def function_name():
    statement
```

Syntax of Identifiers

Identifiers follow specific naming rules. General syntax:

```
identifier_name = value
```

Examples:

```
name = "Shubham"
age = 20
total_marks = 85
```

- ## Question

Q1. What are Keywords in Python?

Keywords in Python are reserved words that have special meanings and are used to define the structure and syntax of the program. These words are predefined by Python and cannot be used as variable names, function names, or identifiers. Examples include if, else, while, for, True, and False. Keywords help control the flow of execution and define logical operations in a program.

Examples of Keywords:

if, else, while, for, True, False

Code:

```
if True:
    print("This is a keyword example")
```

Q2. What are Identifiers in Python?

Identifiers are the names given to variables, functions, classes, or any other objects in a program. They are user-defined and help identify different elements in the code. For example, age, totalMarks, and calculateSum are identifiers. Identifiers should be meaningful and follow naming rules so that the code becomes readable and easy to understand for programmers.

Examples:

age, totalMarks, calculateSum

Example Code:

```
age = 20
totalMarks = 95
print(age, totalMarks)
```

Q3. Rules for Naming Identifiers

- Must start with letter or underscore
- Cannot start with number
- Cannot use keywords

- No special characters
- Case-sensitive

Valid:
name = "Shubham"
_age = 20

Invalid:
1name = "Error"
@value = 50
class = 10

Q4. What are the Characteristics of Keywords?

- Have **fixed meaning**
- Cannot be changed or redefined
- Mostly written in lowercase
- Used for logic, loops, conditions

Example:

```
for i in range(3):
    print(i)
```

Q5. What are the Characteristics of Identifiers?

1. Identifiers are user-defined names.
2. They should be meaningful and descriptive.
3. Identifiers are case-sensitive.
4. They can be short or long (vary in length).
5. They cannot contain spaces.
6. They cannot contain special characters.
7. Good identifiers improve readability and maintenance of code.

Q6. Difference Between Keywords and Identifiers

Keywords	Identifiers
Reserved words	User-defined names
Fixed meaning	No fixed meaning
Cannot be used as names	Used for naming
Define program structure	Store data
Example: if, for	Example: name, marks

Q7. Can Keywords be Used as Identifiers? Explain with Example

- Keywords are reserved words in Python.
- They have fixed meaning and are used for specific purposes.
- Keywords cannot be used as identifiers (variable names).
- Using a keyword as an identifier will give a syntax error.
- Always use a valid identifier instead of a keyword.

Example

```
if = 10      # invalid (keyword used)

value = 10   # valid identifier
```

Q8. Examples of Keywords and Identifiers

Examples of Keywords in Python

Keywords are reserved words, so they have fixed meaning and cannot be used as names.

Common Keywords Examples:

if

else

elif

while

for

break

continue

def

class

True, False, None

Example Code:

```
if True:
    print("This is keyword example")

for i in range(3):
    print(i)
```

Examples of Identifiers in Python

Identifiers are user-defined names used for variables, functions, etc.

Valid Identifiers Examples:

name

age

total_marks

studentName

_value

sum1

Example Code:

```
name = "Shubham"
age = 20
total_marks = 85

def calculate_sum(a, b):
    return a + b
```

Q9. What is Case Sensitivity in Identifiers?

Case sensitivity means that **uppercase (A–Z) and lowercase (a–z) letters are treated as different** in identifiers. In Python, identifiers are **case-sensitive**, so even a small change in letter case creates a completely different name. This is important because using the wrong case can lead to errors or unexpected results in a program.

For example, Name, name, and NAME are considered **three different identifiers** in Python. Programmers must be careful while writing variable names and follow consistent naming styles to avoid confusion and bugs in the code.

Example:

```
Name = "Shubham"
name = "Rahul"

print(Name)  # Output: Shubham
print(name)  # Output: Rahul
```

Q10. Best Practices for Naming Identifiers

1. **Use Meaningful Names**
   Choose names that clearly describe the purpose of the variable.
2. **Keep Names Short and Descriptive**
   Identifiers should be short but still clearly explain their use.
3. **Use Lowercase with Underscore**
   Write identifiers in lowercase and use underscores for better readability.
4. **Avoid Single-Letter Names**
   Do not use single-letter names unless absolutely necessary.
5. **Follow Consistent Style**
   Use the same naming pattern throughout the program.
6. **Write Clean and Readable Code**
   Proper naming helps make the code easy to understand and maintain.